Unwrapping the mind:

knowledge, cognition, and AI

About statistics and hierarchies in language and beyond

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Chair of Psycholinguistics, Anglistik I

What will we talk about today – and why?

LANGUAGE

- What is language? How does it diff[illegible]mal communication or AI?
- How is language repr[illegible]nd brain? How do we know?
- How is language ac[illegible]t mechanisms shape language learning?

DISCLAIMER:
Very selective discussion!

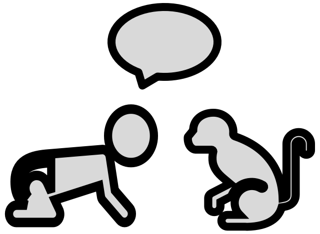
The context: Psycholinguistics

Linguistic theories describe linguistics structures and units (words, phrases, ...), how they are build (grammatical rules) and their meaning (semantics).

Psycholinguistic theories…

- examine the psychological plausability of linguistic units and processes.
- describe the cognitive representation of linguistic knowledge and the processes underlying language behavior and learning.
- are grounded in quantitative and experimental methods that provide empirical evidence for representations and processes that support human language.

Why is psycholinguistics relevant (in this lecture series and more generally)?



Language is one of the key features that makes as human. Psycholinguistics helps us to understand which cognitive representations and processes allow us to learn and use language.

Theoretical contribution

- Assessing the relevance and adequacy of linguistic (language as a system) and psychological (language as cognitive process) theories

Applied contribution

- Diagnosis and therapy of atypical speech, localization of language in the brain (brain surgery), second language learning didactics, human-computer interfaces,....

Before we dive into the topic, think a bit about language, the brain and learning

2 MINUTES BRAINSTORMING

What (type of) information do we need to process to understand language?

What do you know about language processing in the brain?

What are your beliefs about language learning?

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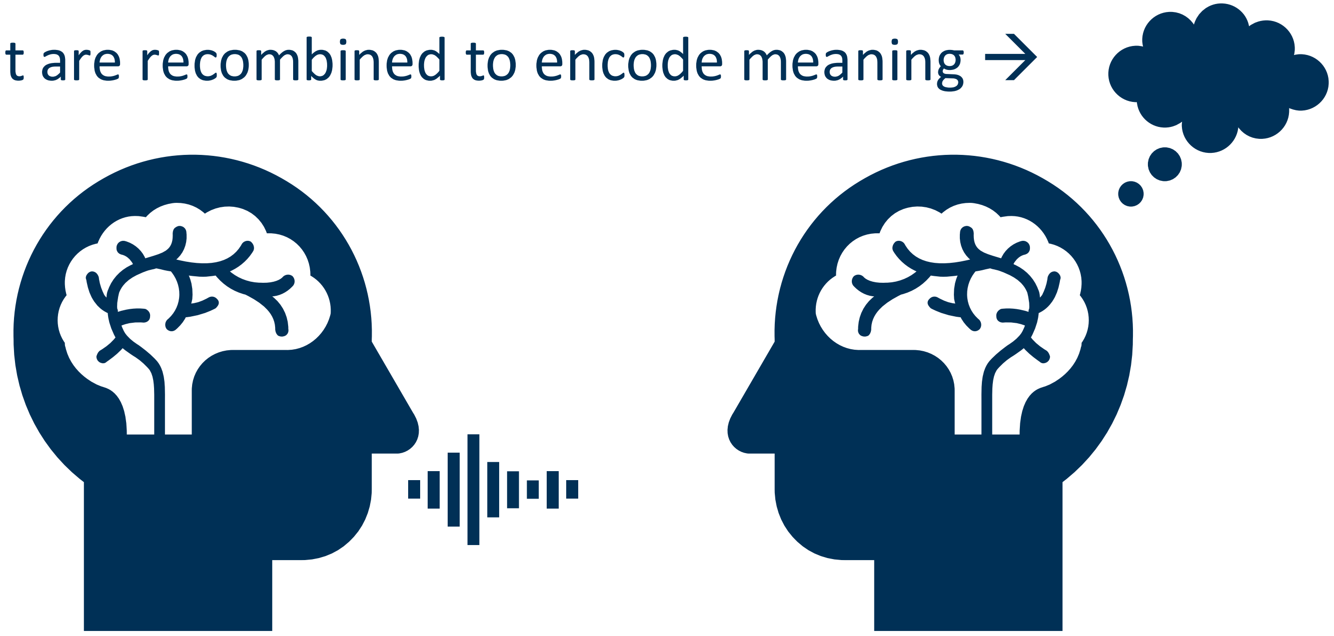
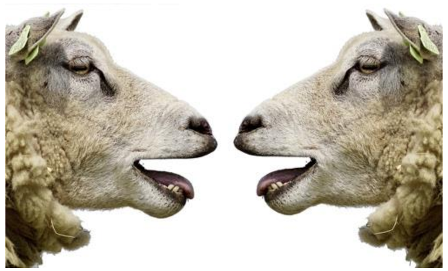
Language as a system

- Multimodal social behavior → COMMUNICATIVE FUNCTION (that also implies inferential reasoning about communicative goals)
- Limited inventory of signs that are recombined to encode meaning → SYMBOLIC FUNCTION



Is human language special? YES!



ANIMALS COMMUNICATE

- Emotion expression (→ tail wagging)
- Limited inventory (→ bee dance)
- Tied to here & now (→ warning calls)
- Linear sequence of signs

HUMANS TALK

- Arbitrary symbols (→ typological differences)
- Expandable inventory (→ new words)
- Representation of abstract (or even counterfactual/impossible) meaning
- Recursive, hierachical sequence of signs

How does language work?

the

park

child

play

laughs

[dog] [[ball] [ ]]

[who are friends]

loudly

and

in

How does language work?

LANGUAGE IS LIKE LEGO!

- Our MENTAL LEXICON contains a myriad of little bricks with specific functional (e.g., plural marking) or lexical (e.g., content nouns) meaning

- We can (re-)assemble the bricks to create new words and sentences according to our language‘s GRAMMAR.

- Most of our COMPETENCE is implicit and unconscious, but this does not limit our PERFORMANCE.

Levels of linguistic information

„Tom writes a letter to his grandma. He wants to thank her for the book.“

Phonology: Recognize sounds/letters and their combination

Lexicon: Identify words and access representation in the mental lexicon

Morphology: Decompose words and isolate grammatical markers

Syntax: Create syntactic frame to understand underlying structure

Semantics: Compose meaning from structure and content

Pragmatics: Bind meaning to discourse and world knowledge

What will we talk about today

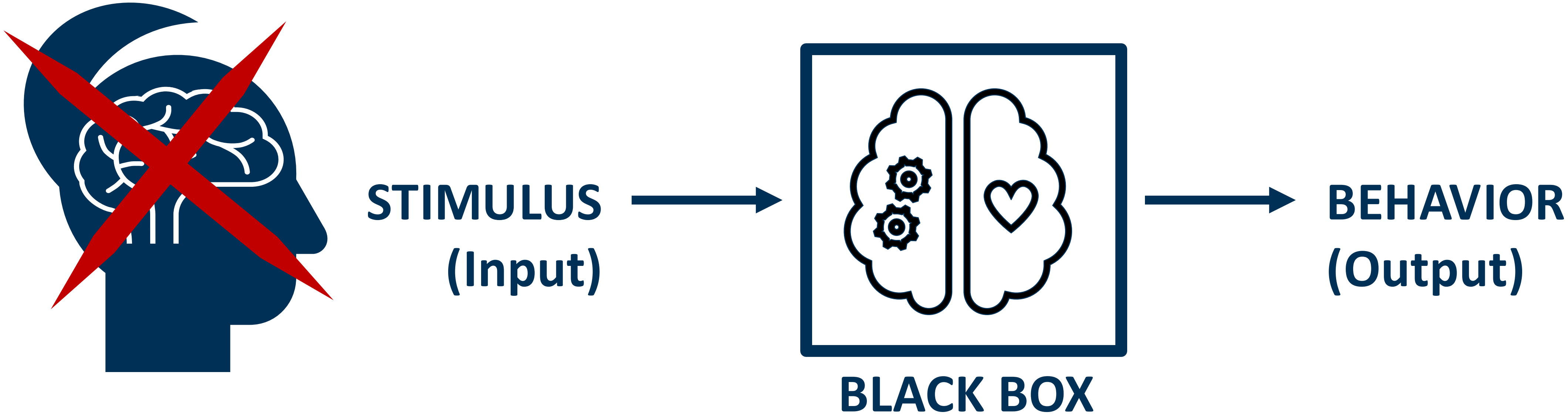
LANGUAGE

- What is language? How does it differ from animal communication or AI?
- How is language represented in the mind and brain? How do we know?
- How is language acquired? What mechanisms shape language learning?

How do we know?

Fundamental problem of psychology:

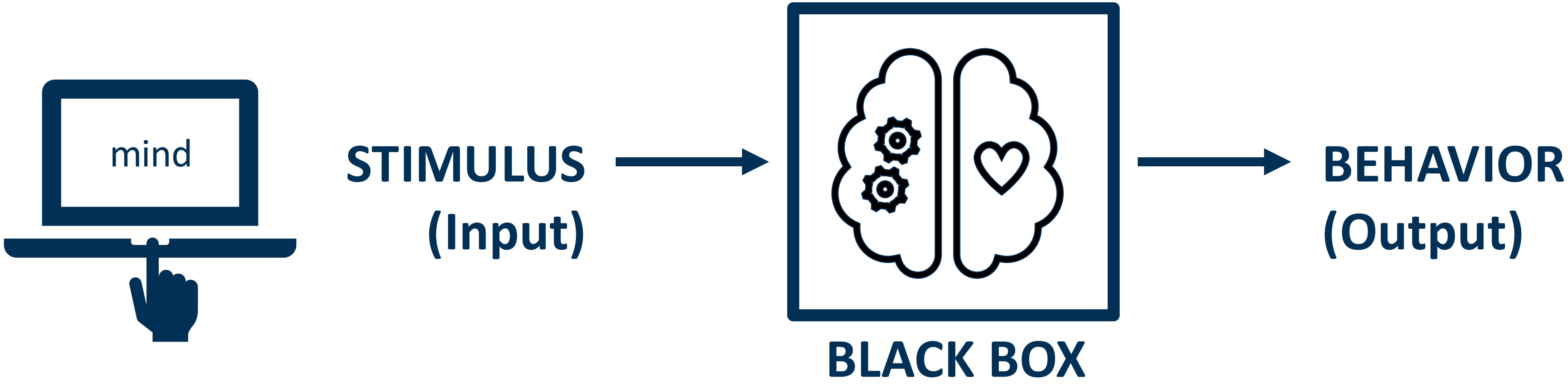
MENTAL PROCESSES AND REPRESENTATIONS ARE NOT DIRECTLY OBSERVABLE.



How do we know?

Experiments help us to draw conclusions:

We systematically manipulate (tiny aspects of) the input (e.g., word frequency) and measure behavioral variables that reflect language processing (e.g., reaction times in word recognition).



How do we know?

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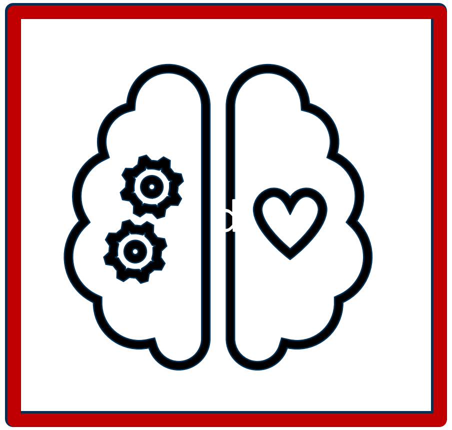
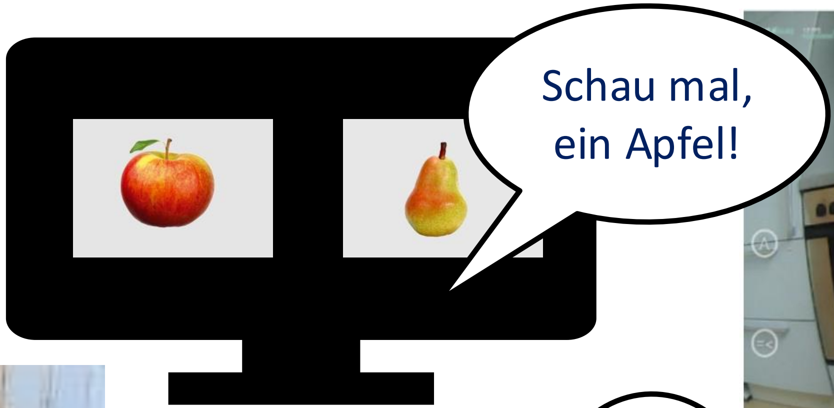
Experiments help us to draw conclusions:

In (preverbal) children we need implicit measures that avoid the need for overt responses. In adults, they can also help to assess the PROCESS rather than just the OUTCOME of language perception/production.

STIMULUS (Input) →  → **BEHAVIOR (Output)**

BLACK BOX

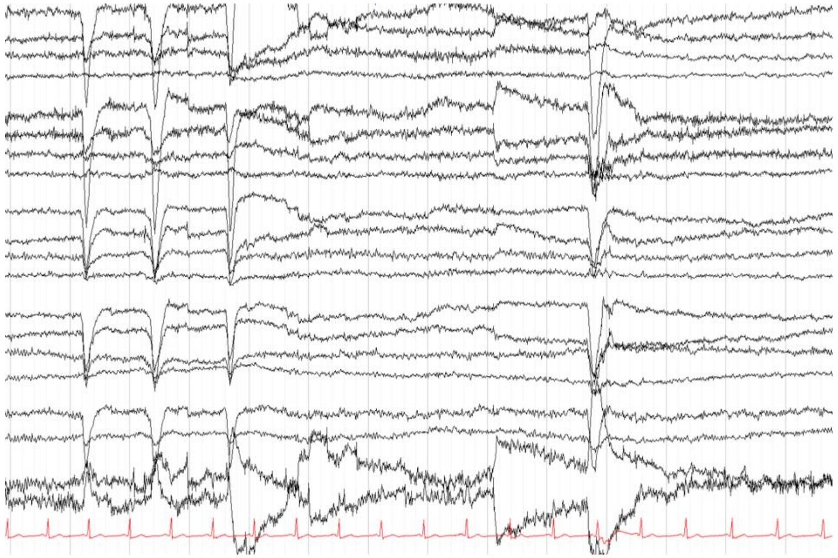
Eye gaze (and pupil dilation) as a measure of attention and processing



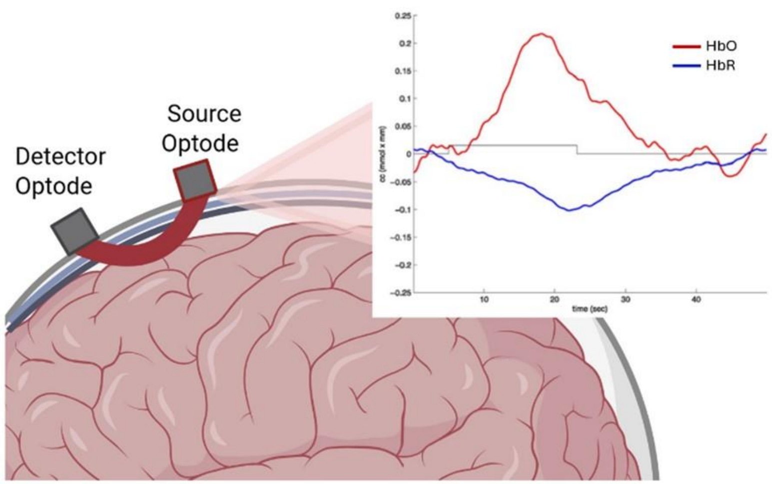
- Fixation latency
- Fixation duration
- Pupil size

Neural measures of processing components, their timing and location

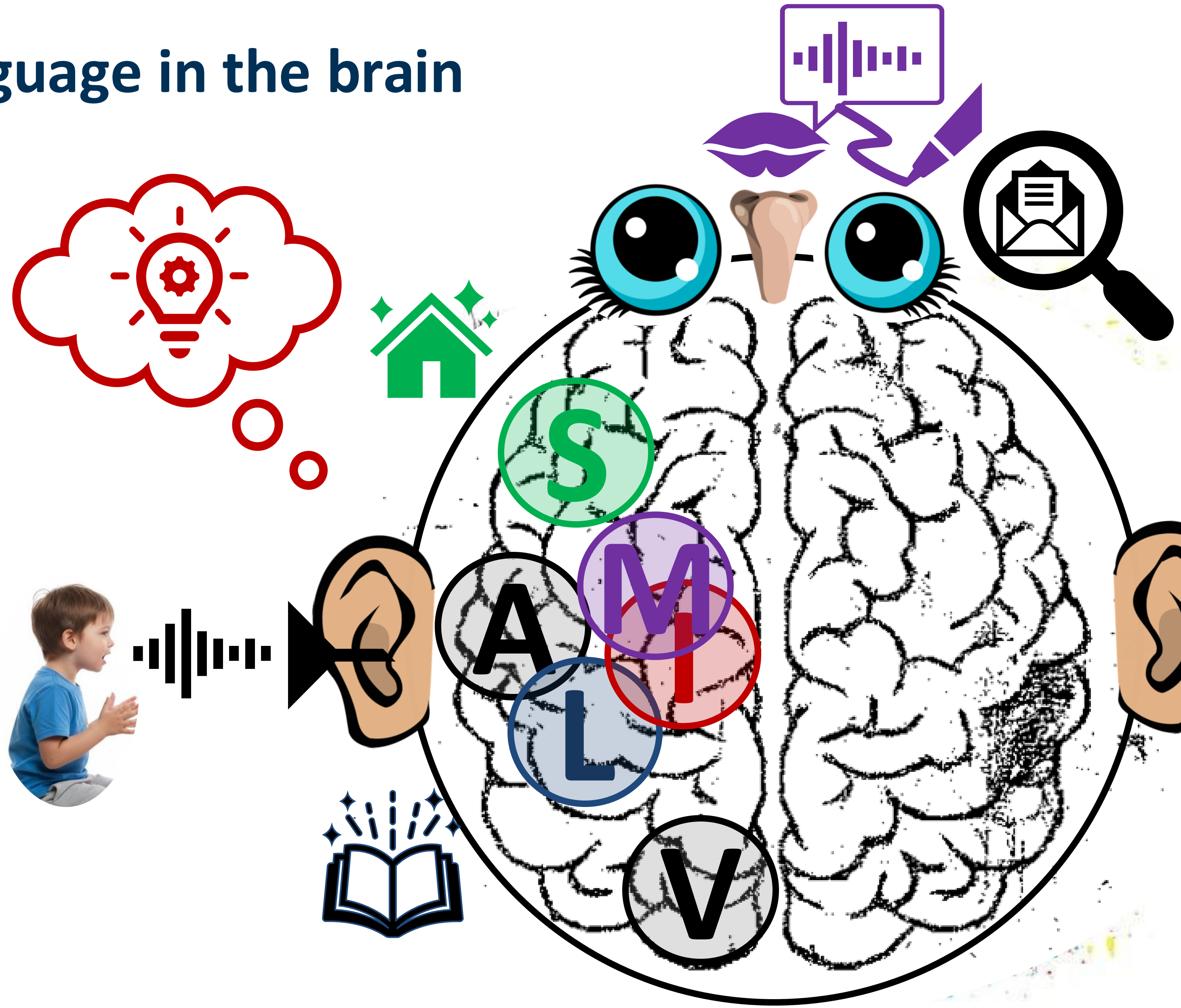
EEG = Electroencephalography
ERP = event-related component



fNIRS = functional near-infrared spectroscopy



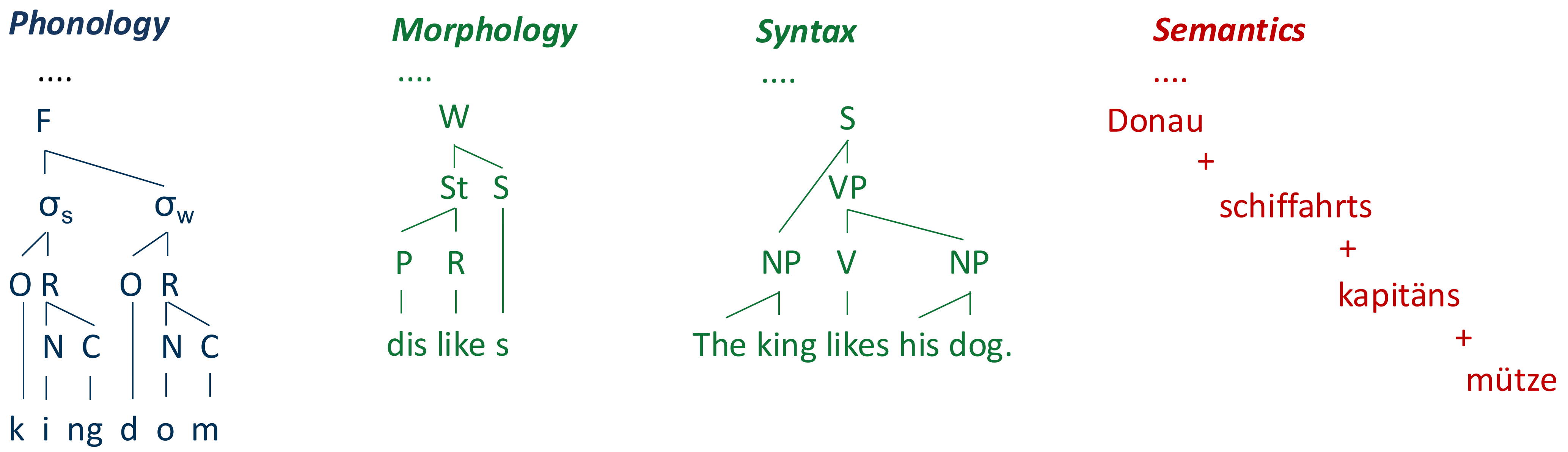
Language in the brain



Right hemisphere
- Prosody/Intonation
- Emotion
- Non-literal meaning
- Formulaic language
- …

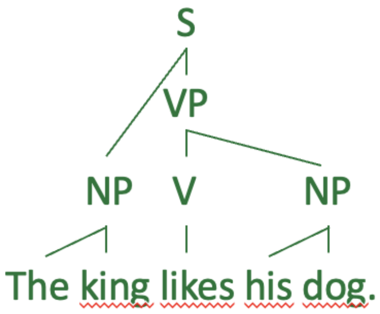
Hierarchical structure in language

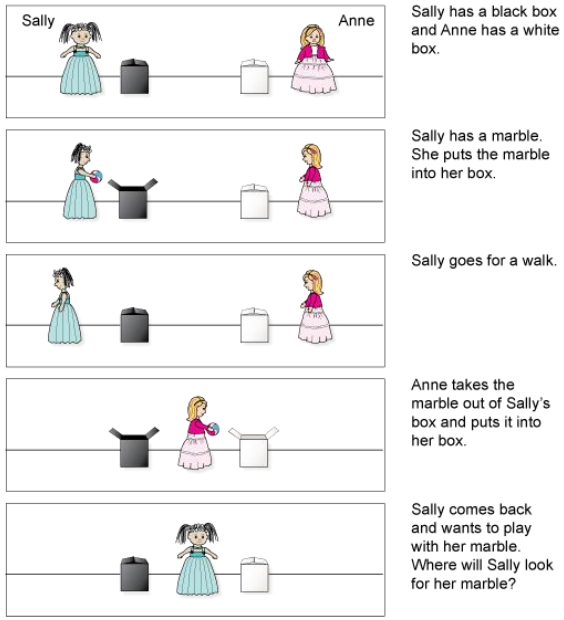
Key elements of linguistic structure are ***hierarchy***, ***recursivity*** and ***compositionality***.

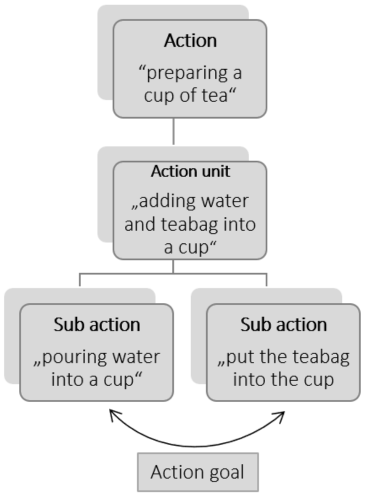


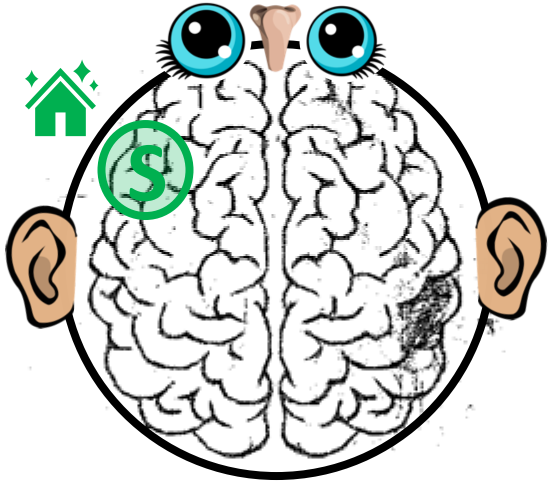
Hierarchical structure in language - and beyond

It has been argued that action and Theory of Mind rely on similar structural hierarchies as language, i.e. that it is a **domain-general cognitive feature**.









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Hierarchical structure in language - and beyond

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Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind?

Series of studies that compare processing (and its development) across domains

- correlating understanding of differently complex sentences and actions, Theory of Mind, problem solving, executive functions and working memory (behavioral)
- comparing processing of language and daily actions (EEG)
- assessing brain activation for language, action and Theory of Mind (fNIRS)
- correlating behavioral performance with brain maturation (structural MRI)

Lea Härms

Laura Maffongelli

Angela Friederici

Markus Paulus

Hierarchical structure in language - and beyond

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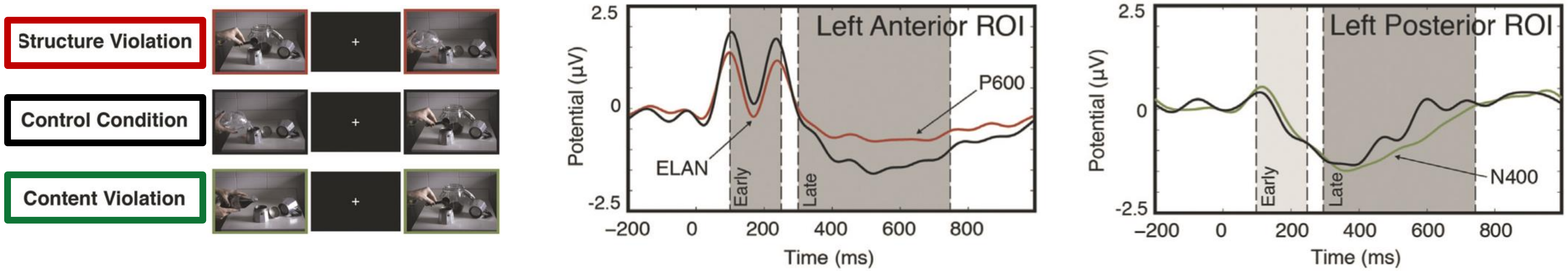
Laura Maffongelli

Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from EEG

Structural processing in language is associated with an early left anterior negativity (ELAN) and a late posterior positivity (P600). Similar components have been found for violations in the order of action steps (Maffongelli et al., 2015)



Hierarchical structure in language - and beyond

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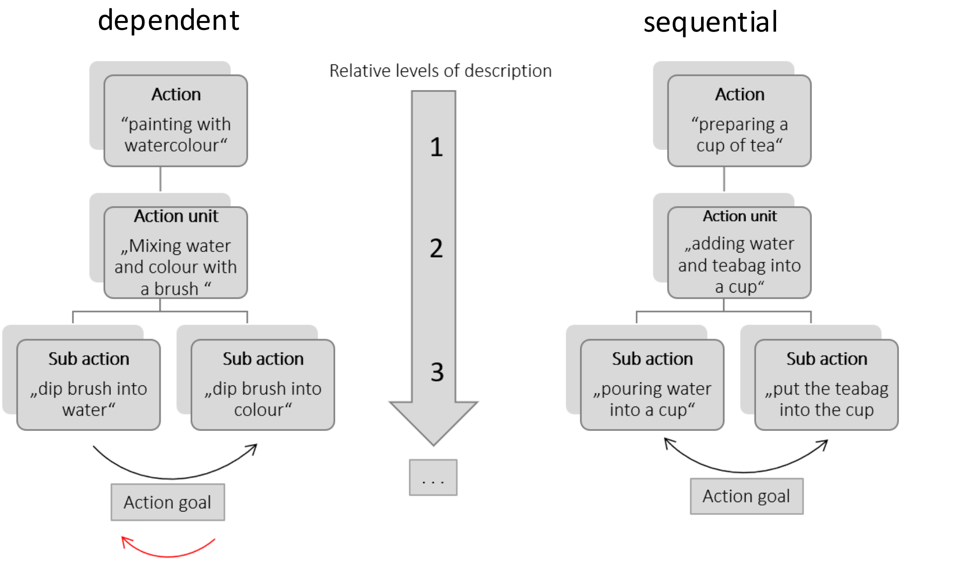
Laura Maffongelli

Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from EEG



Stimuli: 20 sequences of each type (80 total)
Participants: 22 young adults, 22 preschoolers

(Maffongelli et al., in prep)

Hierarchical structure in language - and beyond

Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from EEG ✗

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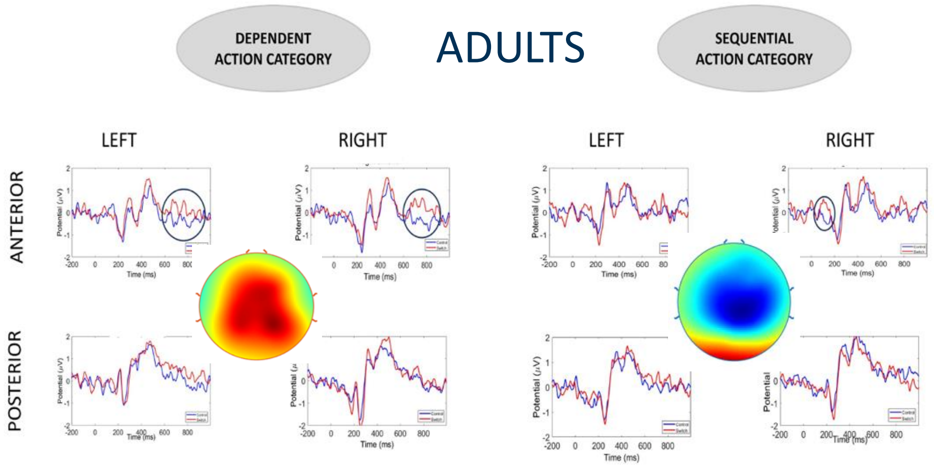
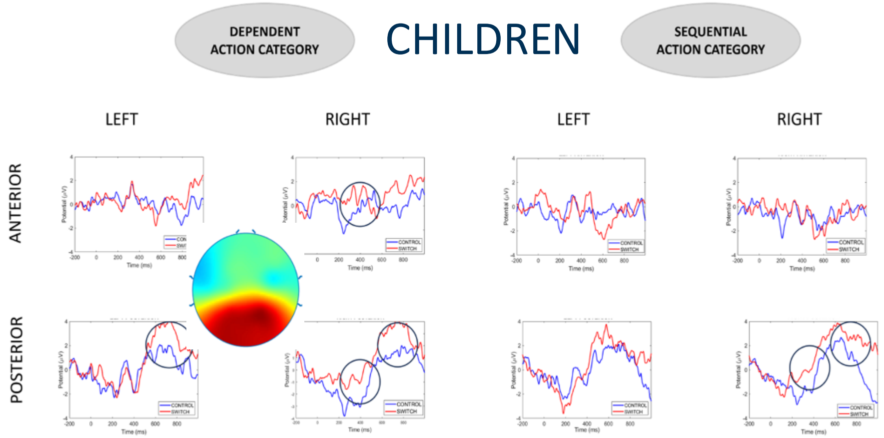
Laura Maffongelli





→ late positivity modulated by stimulus complexity

→ developmental differences between children & adults (also behavioral differences in accuracy!)

Hierarchical structure in language - and beyond

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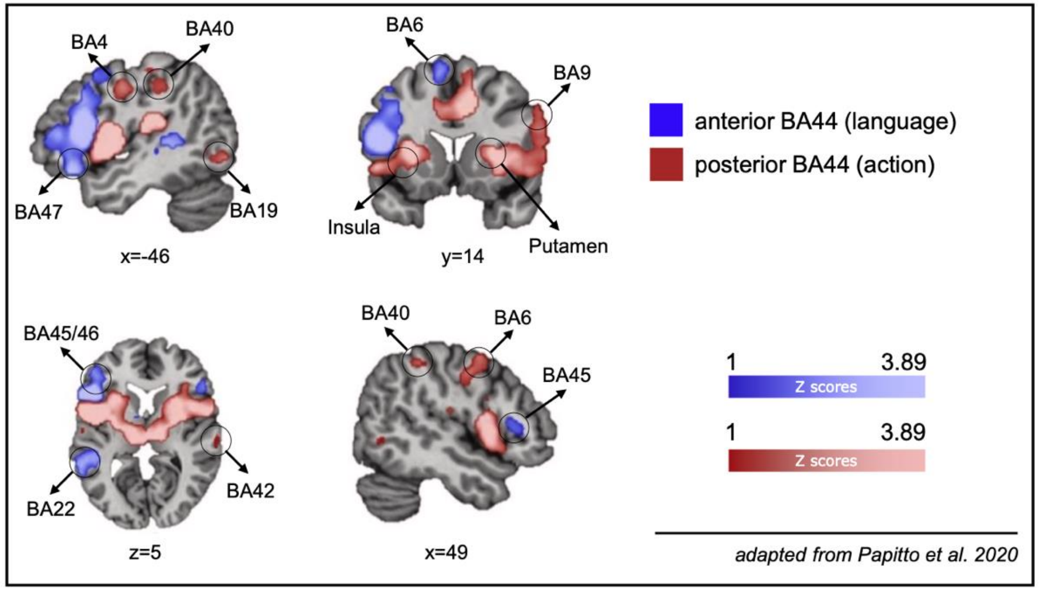
Lea Härms

Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from fNIRS

Structural processing in language recruits Broca‘s area (BA 44/45) in left inferior frontal gyrus. Adult action processing has been associated with partially overlapping but distinct (BA 44) and adjacent (BA 6) bilateral areas (Zaccarella et al., 2021).



Hierarchical structure in language - and beyond

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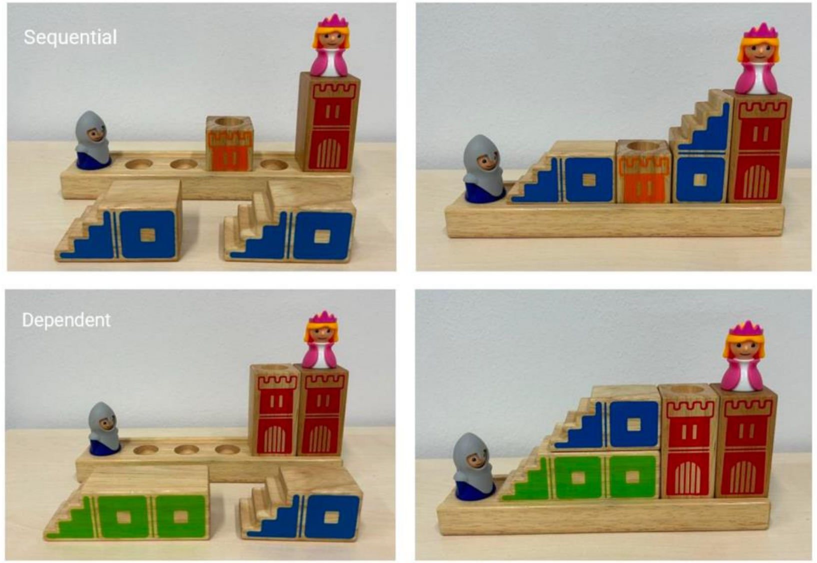
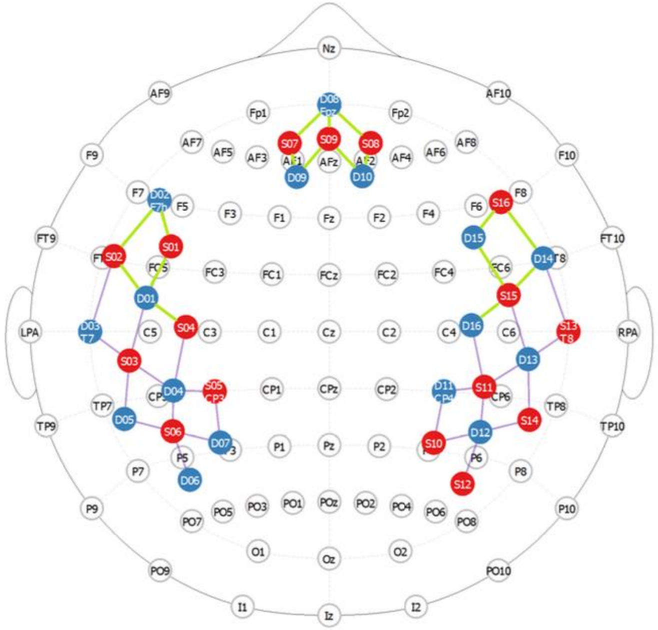
Lea Härms

Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from fNIRS





Stimuli:
6 levels of each type
(12 total)

Participants:
36 young adults
72 preschoolers

(Härms et al., in prep)

Hierarchical structure in language - and beyond

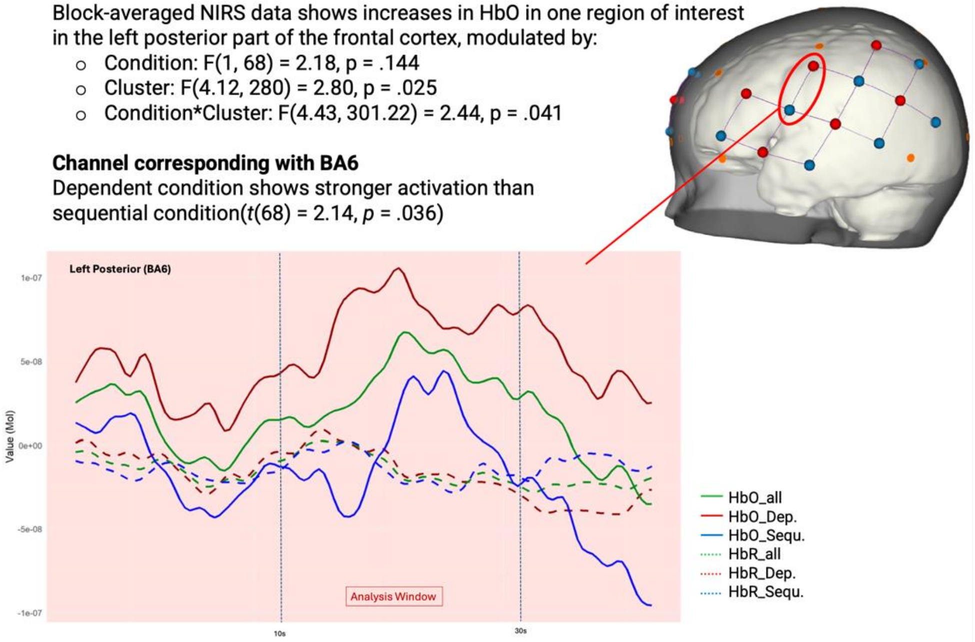
Does **structural processing** rely on shared neural mechanisms across language, action and Theory of Mind? → Evidence from fNIRS ✗

Group	Ø age (range)	Success ...overall	sequential	dependent
All*	60.2 (38-82)	7.18 (0-12)	4.63 (0-7)	2.54 (0-5)
Low perform. (n = 32)	56.5 (37-80)	3.30 (0-5)	3.41 (0-6)	1.35 (0-4)
High perform. (n = 36)	63.2 (51-80)	9.31 (7-12)	5.71 (4-7)	3.60 (1-5)

Block-averaged NIRS data shows increases in HbO in one region of interest in the left posterior part of the frontal cortex, modulated by:

- Condition: $F(1, 68) = 2.18, p = .144$
- Cluster: $F(4.12, 280) = 2.80, p = .025$
- Condition*Cluster: $F(4.43, 301.22) = 2.44, p = .041$

Channel corresponding with BA6
Dependent condition shows stronger activation than sequential condition($t(68) = 2.14, p = .036$)



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(Härms et al., in prep)

Hierarchical structure in language - and beyond?

DFG FOR 2253

While hierarchical processing behaviorally correlates across domains and emerges at similar times in the preschool years, neural measures provide only limited evidence for a shared underlying processing mechanism.

What will we talk about today

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- What is language? How does it differ from animal communication or AI?
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What needs to be learned?

GRAMMAR
Set of rules how elements may be combined

LEXICON
Storage of form-meaning pairs

flaʊə

/g/ -/k/
dog – dock
**$_{\sigma}$[lg – $_{\sigma}$[gl*
**lgad – glad*

Phonology

Sound system

they play
*he play**s***

Morphology

Grammatical markers on words

Mummy likes flowers.
**Mummy flowers likes.*

Syntax

Rules on word combinations

Semantics

Assigning meaning

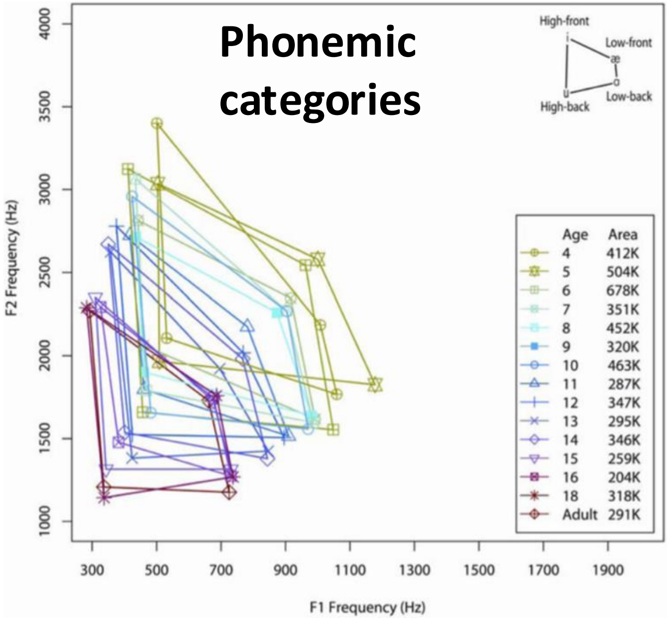
When may I ask a question?

Pragmatics

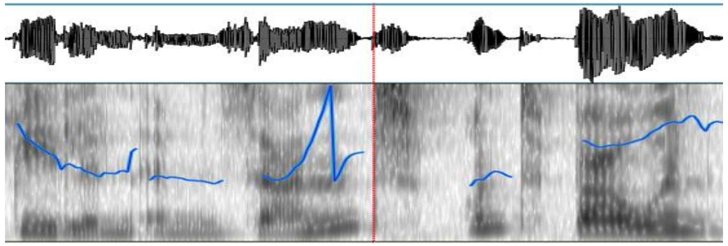
Conversational rules

What is the problem?

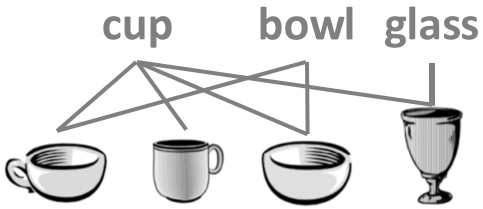
The input is noisy and does not present clear-cut categories. Children have to deduce abstract representations from a broad range of overlapping concrete exemplars.



Word segmentation



Word meaning



Morphological regularities

Hund *dog* – Hunde *dogs*
Puppe *doll* – Puppen *dolls*
Kater *male cat* – Kater *male cats*
Tochter *daugther* – Töchter *daugthers*

Word order

Le petit prince (DAN) the little prince
En infant terrible (DNA) a terrible child
.....

Ressources that help children to solve the learning problem include...

Innate biases

Predispositions that help children to attend to relevant input, e.g. preference for socially relevant stimuli (faces, language, infant-directed communication, pitch countours, …)

Scaffolding

Adult behavior that supports language learning (infant-directed speech, imitation, turn-taking, …)

Cognition

General cognitive mechanisms supporting learning (categorical perception, associative learning, fast mapping, statstical learning, …)

Principles

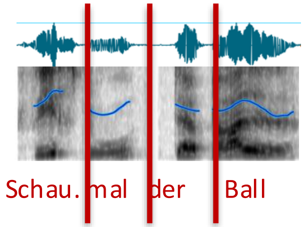
Domain-specific linguistic principles that facilitate learning (mutual exclusivity, Gricean maximes, universal grammar?)

Are babies like little machines when it comes to statistical computations?

Machine learning

→ finding a predictive function that maps input/output pairs based on a (limited) data set

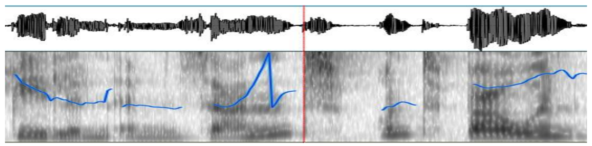
→ often supervised learning with training on an annotated set and testing prediction on an untrained set



Human learning (domain-general)

→ detect statistical regularities in the world to categorize and learn about the environment

→ mostly unsupervised, implicit learning (or reinforcement learning) with opaque input from which categories have to be derived

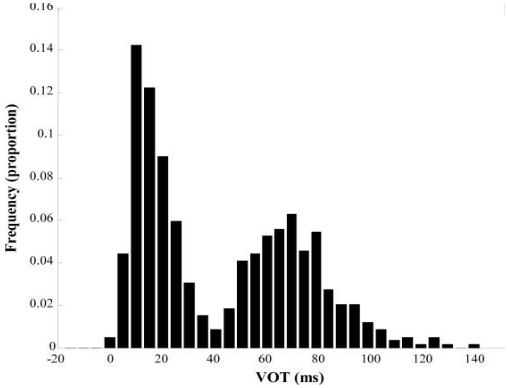


→ ?

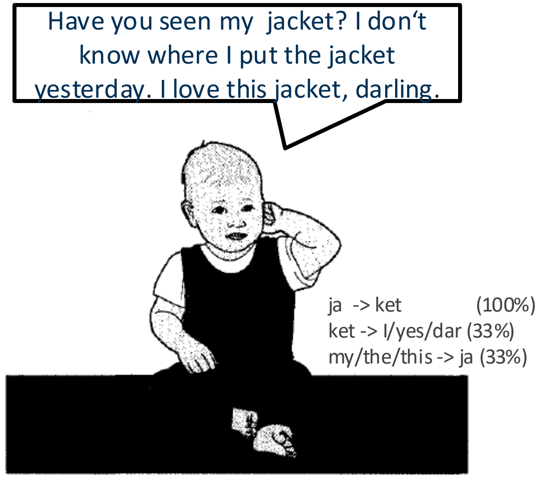
Is language a predictable system?

Linguistic forms are associated with specific frequency and co-occurrence information to which infants are sensitive and that may help to infer/learn linguistic categories.

Distributional information
Example: phonemic categories

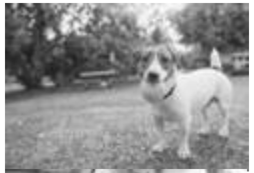


Transitional probabilities
Example: speech segmentation



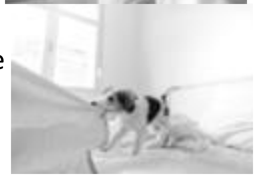
Co-occurrence information
Example: word learning

The **dog** fetched the **ball**



The **child** throws the **ball**

The **dog** tugs at the blanket.



Non-adjacent dependencies
Example: tense marking

Past tense:

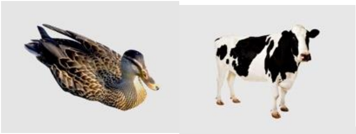
He **has** push**ed**
She **has** pull**ed**
She **has** mov**ed**
He **has** call**ed**
....

*-> S/he **has** X-**ed***

How does it work in the wild?

Psycholinguistic experiments take place in controlled laboratory settings and use limited stimulus material, but children need to solve the learning problem in a complex environment.

LABORATORY ⟷ REAL WORLD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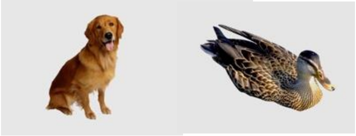
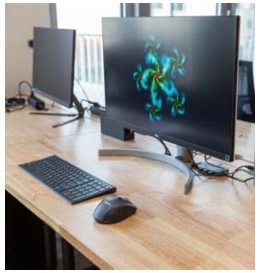
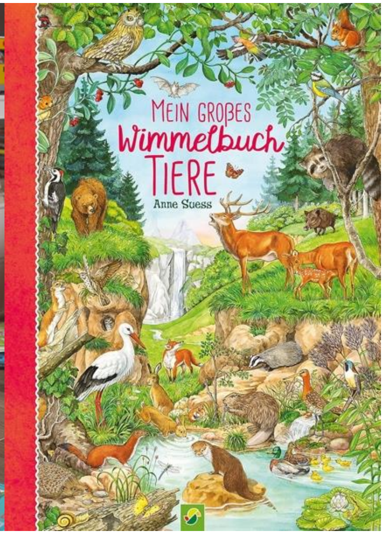
Momp


Foffel


Lark





Shared book reading as a test case

Shared book reading is a common behavior that provides rich opportunity for interaction and learning. Quality and quantity of reading is associated with vocabulary growth, grammar development and later literacy.

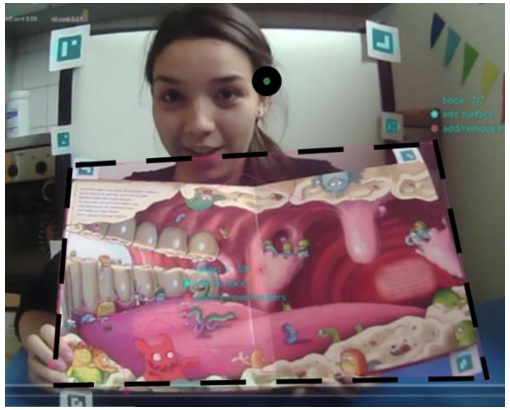
- Positive atmosphere
- Focused attention
- Child-led communication
- Dialogic conversation
- Rich (and partially infrequent) vocabulary
- Complex sentence structure
- ....

(e.g., Mol et al. 2008, Montag et al., 2015)

Shared book reading as a test case

We developed child-friendly eye tracking glasses to monitor attention and learning out of the lab in semi-controlled interactive settings.

Reading (~3.5 min):
- One-to-one setting
- New character and word embedded in scripted story



Participants:

N = 45, Ø 45 mo (2-4 ys)
♀ 23, 16 bilingual

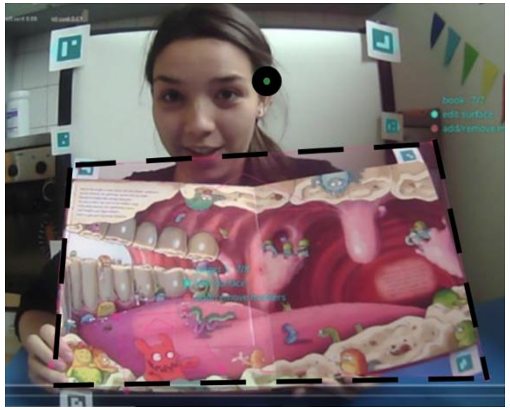
(Altvater-Mackensen, 2021; in prep)

Shared book reading as a test case

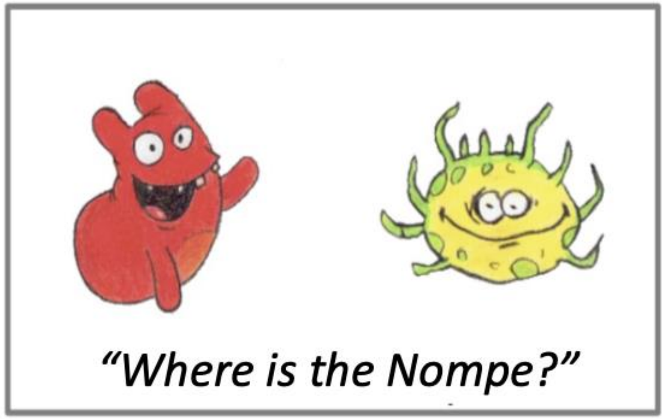
We hypothesized that attention distribution during learning is associated with children‘s word learning.

Attention to speaker
- Visual speech cues (mouth)
  → phonological form of word

Attention to book
- Statistical co-occurrence of specific character and word
  → semantic referent of word



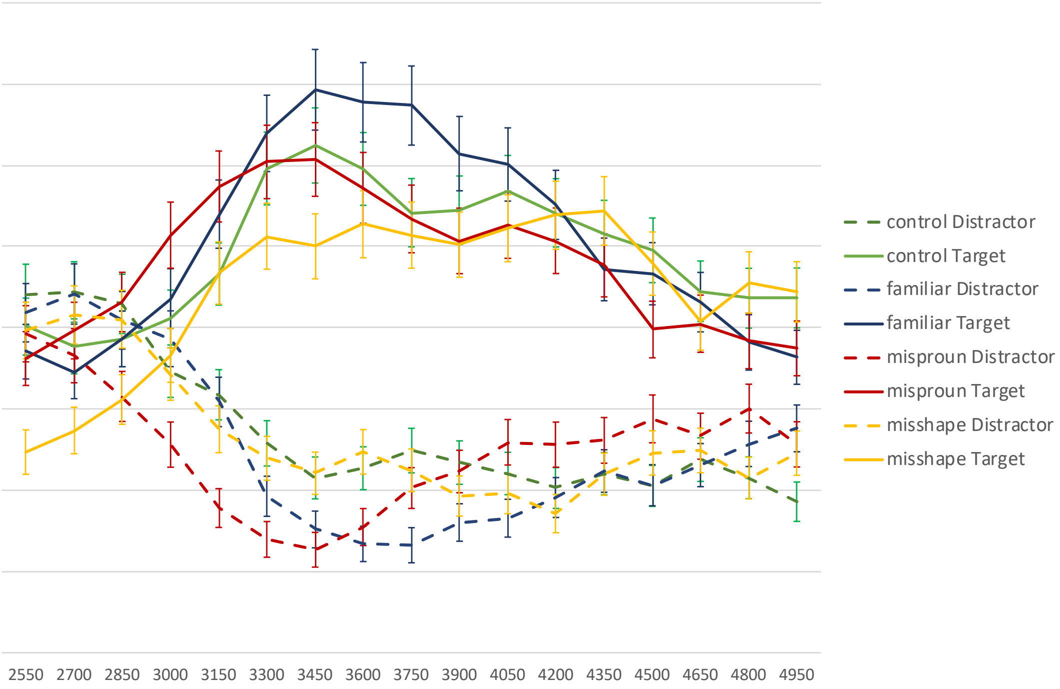
Nompe

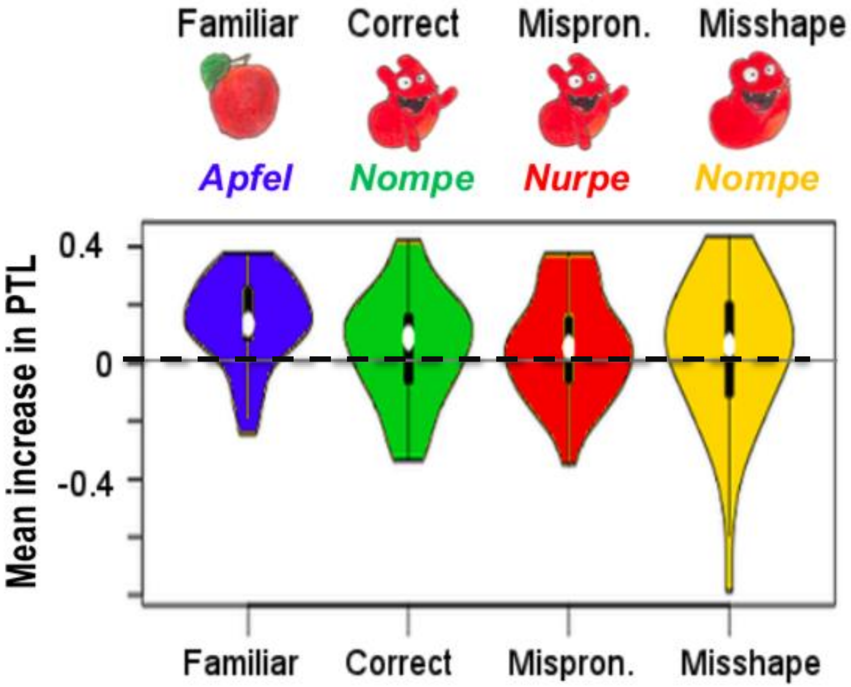


IPL-task (32 trials):
- Familiar and novel words
- *Nompe*-trials with changes in word and referent form

Shared book reading as a test case

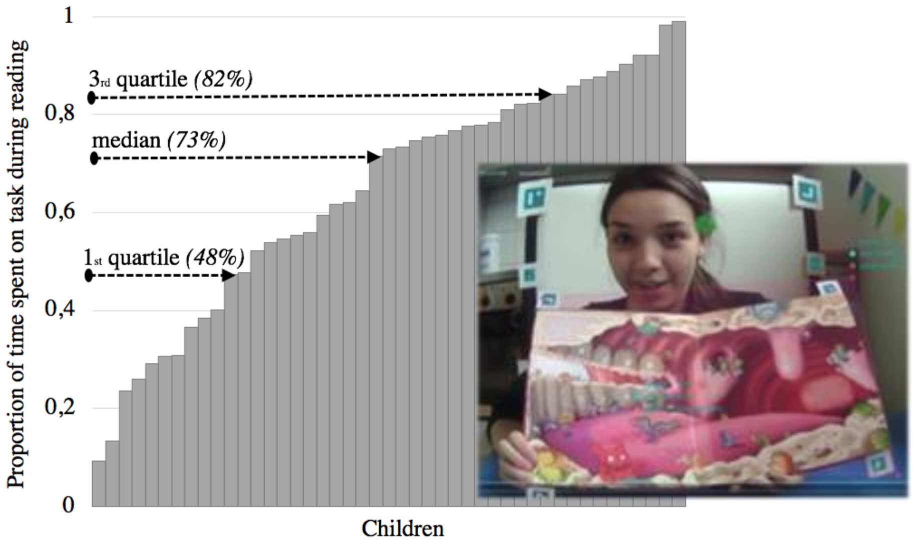
Children are sensitive to mispronunciations and changes in form → they learned the novel word during reading, i.e. in a “natural“ cross-situational word learning task.





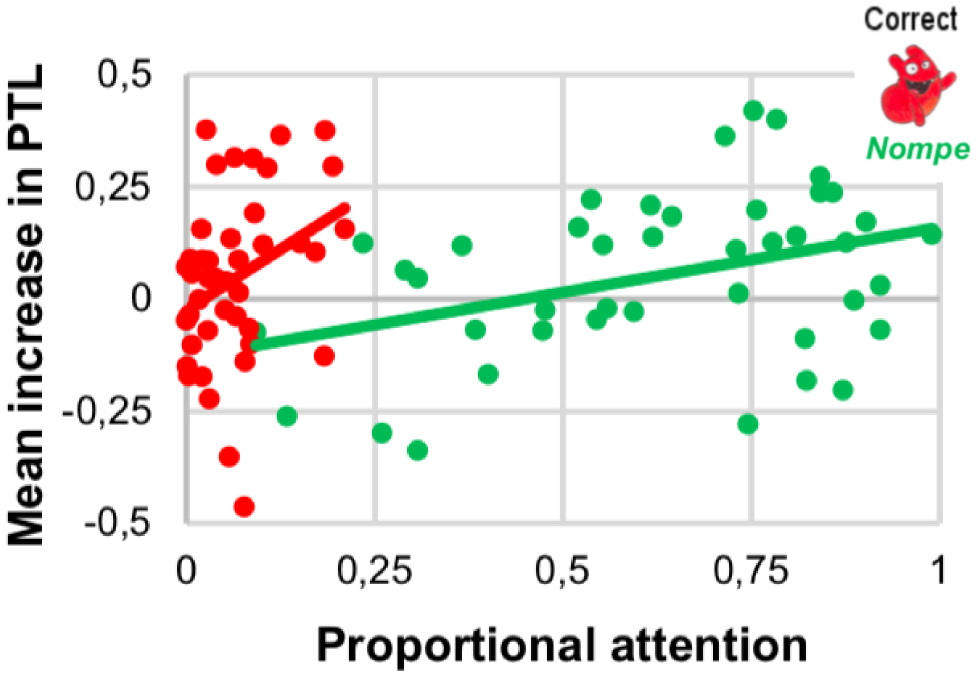
Shared book reading as a test case

Even though children are overall very attentive, distribution of attention during reading shows large individual differences (associated with learning).



Attention to **reading** Ø 62% (range 9-99)

→ PTL in **correct trials** (r_s = .35*)



Shared book reading as a test case

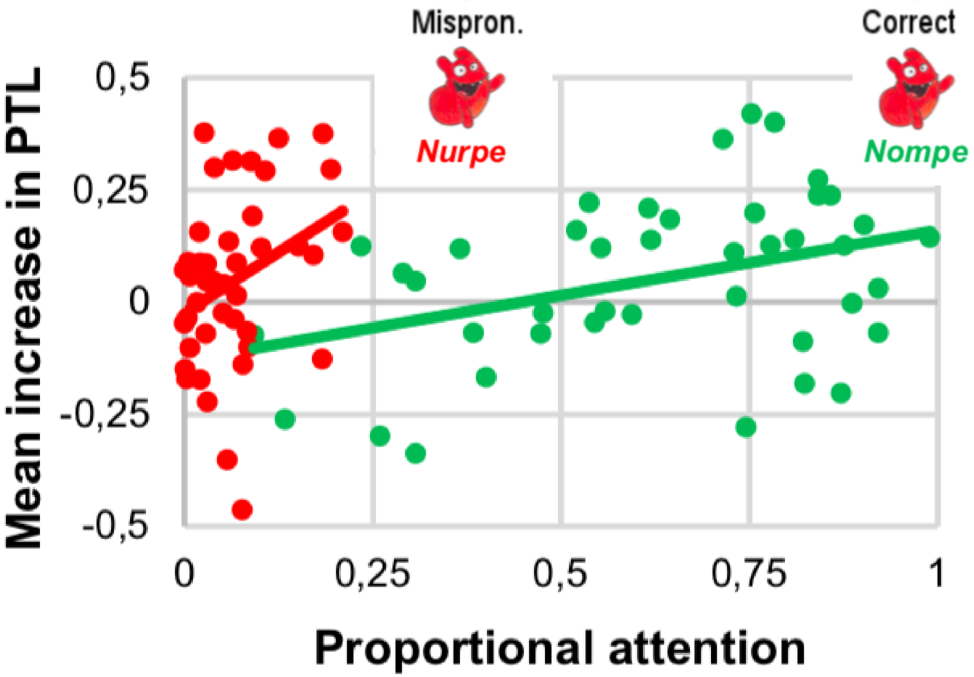
Even though children are overall very attentive, distribution of attention during reading shows large individual differences (associated with learning).

Visual speech
→ **phonological encoding**

Attention to **speaker** (Ø 7% , range 0-27) correlates with PTL in **mispronounced trials** (r_s = .39**)

Referential information
→ **semantic encoding**

NO association between attention to **book** and PTL in **misshaped trials** (p >.1)



Shared book reading as a test case

Children track co-occurrences of words and objects in natural book reading, i.e. they learn from crowded input and without explicit instruction. Distribution of attention during reading can predict what type of information is memorized in word learning.

Outlook:

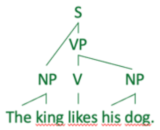
- Controlled word learning study to assess recruitment and impact of different cues (speaker‘s mouth and eyes, referent object) on word learning
- Correlational study assessing reading quality and language outcomes

What have we talked about today

LANGUAGE

- What is language? How does it differ from animal communication or AI?

  - Language systematically combines abstract symbols according to rules.
  - In contrast to animal communication, it is hierarchical, recursive and compositional.
- How is language represented in the mind and brain? How do we know?



  - Behavioral and neurocognitive measures allow conclusions on mental representation.
  - Even though related, language differs from other cognitive domains.
- How is language acquired? What mechanisms shape language learning?

  - Children use a wide array of language-specific and domain-general mechanisms.
  - Quantity and quality of input is critical and are partially (self-)shaped by children.

Questions?

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More information on the lab (in German)
https://www.phil.uni-mannheim.de/en/anglistik/abteilungen/anglistik-i/forschung/wortakrobaten/